Antonio Coppola

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Optimization engineer with 4+ years building and deploying decision-support systems for large-scale mobility networks, combining deep expertise in combinatorial optimization and AI/ML, with a track record of measurable real-world impact. I thrive at the intersection of rigorous problem-solving and direct stakeholder engagement. I am looking for a role that demands both technical ownership and client-facing responsibility.

EXPERIENCE

Optimization Engineer (Ph.D. Candidate, Mobility AI) 2022–2026
Technical University of Munich | supervised by Prof. Maximilian Schiffer

- Conceived and implemented simulation frameworks for congestion-aware autonomous taxi routing.
- Built advanced optimization solvers utilizing matheuristics, metaheuristics, and ML-guided solvers (GLMs, MLPs, GNNs).
- Achieved up to 35% reduction in traffic congestion delays across large-scale urban networks under real-world demand scenarios.
- Developed end-to-end simulation, optimization, and visualization pipelines in Python, Julia, and C++ using Git and CI/CD.
- Published in top-tier peer-reviewed journals; 2 manuscripts currently under revision.
- Managed international research collaborations across TUM, Polytechnique Montréal, and Imperial College London.
- Presented at 5 operations research and transportation conferences across Europe and North America.
- Supervised Master's theses and served as teaching assistant.

Algorithm Engineer — Large-Scale Transportation Optimization 2025
Polytechnique Montréal | hosted by Prof. Michel Gendreau

- Collaborated with a world-leading researcher in transportation science on large-scale optimization problems and metaheuristic design.
- Implemented and benchmarked state-of-the-art large neighborhood search algorithms for real-world vehicle routing.
- Engaged in international operations research collaborations, including conference presentations and academic reading groups.
- Produced technical reports and presentations communicating complex optimization results to both academic and applied audiences.

Software Engineer — Transit Optimization & Field Deployment 2024
MCube Cluster (Munich Cluster for the Future of Mobility) | STEAM Project

- Led end-to-end development of a semi-flexible transit optimization system, from algorithm design to full field deployment.
- Collected, processed, and analyzed GPS telemetry data across a 3-day large-scale empirical mobility experiment.
- Designed demand-responsive scheduling algorithms, reducing walking distances by 20% and improving vehicle occupancy by 15%.
- Interfaced directly with industry partners (BMW, SWM, MAN), translating operational needs into algorithmic solutions.

Engineering Intern — Network Dynamical Systems 2021
University of Naples Federico II | supervised by Prof. Franco Garofalo

- Co-authored peer-reviewed publication in *IEEE Control Systems Letters* on controllability of complex network dynamical systems.
- Contributed to conceptualization, visualizations, formal analysis, and manuscript writing for the final publication.

EDUCATION

Ph.D. in Operations Research (Candidate) 2022–2026
Technical University of Munich, Germany
Thesis: *Balanced and Staggered Routing in Autonomous Mobility-on-Demand Systems*

B.Sc. & M.Sc. in Industrial Engineering (110/110 *cum laude*) 2014–2021
University of Naples Federico II, Italy
Thesis: *Reconstruction of the Connectivity of Diffusion Network Processes from Temporal Traces*

PUBLICATIONS

A. Coppola et al. — *Staggered Routing in AMoD Systems*, European Journal of Operational Research, 2025
A. Coppola et al. — *Integrated Balanced and Staggered Routing in AMoD Systems*, under revision, 2026
A. Coppola et al. — *Online Balanced and Staggered Routing in AMoD Systems*, under revision, 2026
C. Ancona, ..., **A. Coppola** et al. — *Partial Controllability of Network Dynamical Systems*, IEEE Control Systems Letters, 2021

COMPETITIVE GRANTS & SCHOLARSHIPS

Bavaria–Québec Mobility Program — BayFOR 2025
Academic Training Program — TUM School of Management 2024
Erasmus+ Program — European Commission 2019

SKILLS

Optimization: LP/IP/MILP, stochastic & network optimization, metaheuristics, matheuristics, Gurobi, CPLEX
ML & AI: Deep learning, GLMs, MLPs, GNNs, feature engineering, predictive analytics; PyTorch, scikit-learn
Engineering: Python, Julia, C++, TypeScript, Docker; Git, CI/CD; data pipelines, large-scale simulation, experimental design
Languages: Italian (native), English (fluent), German (proficient), Portuguese & French (conversational)